

LAB 04

```
import pandas as pd

data = pd.read_csv("training_data.csv")

print("Training Data Loaded:")

print(data)

print("-" * 30)

def find_s_algorithm(data):

    """Implements the Find-S algorithm to find the most specific hypothesis."""

    attributes = data.iloc[:, :-1].values

    target = data.iloc[:, -1].values

    hypothesis = None

    for i in range(len(target)):

        if target[i] == "Yes":

            hypothesis = list(attributes[i])

            break

    if hypothesis is not None:

        for i in range(len(target)):

            if target[i] == "Yes":

                for j in range(len(hypothesis)):

                    if hypothesis[j] != attributes[i][j]:

                        hypothesis[j] = '?'

    return hypothesis
```

```
final_hypothesis = find_s_algorithm(data)
```

```
print("\nFinal Learned Most Specific Hypothesis:")
```

```
print(final_hypothesis)
```